

Mutian (Austin) He

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EDUCATION

Santa Clara University, Leavey School of Business

Santa Clara, US

Master of Science in Information Systems

Dec 2026

- Data Analytics with Python, Cloud Computing, Database Systems, Machine Learning, Deep Learning, NLP

University of Glasgow, School of Computing Science

Glasgow, UK

Bachelor of Science in Computer Science (BSc CS)

Jun 2024

- Data Structures & Algorithms, Database Systems, Software Engineering, Human-Computer Interaction

PRACTICUM

Amazon Web Services (AWS)

AI/LLM Engineer Intern

Jan 2026 – Jun 2026

RiskLens AI – Intelligent Risk Analysis Platform | Python, AWS, LLM/NLP

[Live Demo](#) | [Github](#)

Project Description: Designed and implemented an AWS-based data and LLM Agent-driven analytics platform (serving both B2B analysts and end users), transforming unstructured SEC 10-K filings into structured data. Enabled risk change detection and decision support through multi-dimensional risk scoring and cross-year comparison.

Responsibilities:

- Led system architecture design and core module development; **built data pipelines on AWS** (S3, Textract, Bedrock) to parse, clean, and structure unstructured 10-K filings.
- Designed a hybrid **rule-based + LLM extraction framework**, leveraging prompt engineering to **improve risk factor identification** and implemented cross-year difference detection and financial data parsing.
- Developed an LLM Agent for **multi-dimensional risk scoring** and automated report generation; integrated stock and news data with interactive dashboards to **analyze correlations** between risk factors, market performance, and sentiment changes.

PROJECTS

Image Search System – Multimodal Machine Learning (2026) | PyTorch, CLIP, FAISS

[Live Demo](#) | [GitHub](#)

- Built a **CLIP-based multimodal retrieval system** enabling text-to-image search by projecting images and queries into a shared embedding space.
- Designed an end-to-end retrieval pipeline including **data preprocessing, batch image embedding generation, vector indexing, and Top-K similarity search**.
- Leveraged **FAISS for efficient approximate nearest neighbor (ANN) search**, significantly improving scalability and latency over brute-force retrieval.
- Implemented **evaluation metrics (Recall@1/5/10 and Median Rank)** to quantitatively assess retrieval performance and validate model effectiveness.

Cloud-Based Stock Data Pipeline –Data Engineering (2026) | AWS (S3, Glue, Athena), SQL, Kafka, ETL

[Github](#)

- Built a real-time stock data pipeline to enable **low-latency market data ingestion** and analysis, supporting **near real-time trading insights**.
- Designed a scalable AWS data lake (S3 + Glue) to **centralize fragmented data sources** and improve **data accessibility**.
- Automated streaming ingestion workflows, reducing **manual data processing** and enabling **continuous data availability**.
- Enabled serverless querying with Amazon Athena, allowing **faster exploratory analysis** and reducing **infrastructure overhead**.

Bank Marketing Subscription Prediction – Machine Learning (2026) | Pandas, Scikit-Learn, SMOTE, SHAP

[Github](#)

- Built a machine learning pipeline to **predict customer subscription likelihood**, enabling **more effective targeting in marketing campaigns**.
- Addressed **severe class imbalance (1:9)** using SMOTE and threshold tuning, improving **minority-class recall and campaign reach**.
- Optimized decision threshold to **prioritize high-probability customers**, supporting **more efficient allocation of marketing resources**.
- Applied SHAP to **identify key drivers of customer conversion**, informing **data-driven marketing strategy** (ROC-AUC: 0.80).

SKILLS

Programming: Python, SQL (MySQL, SQLite, PostgreSQL, MongoDB), Java

Data Engineering: Apache Kafka, Apache Airflow, ETL/ELT Pipelines, Streaming Architecture, Data Modeling

Machine Learning / Deep Learning: Scikit-learn, Feature Engineering, Cross-Validation (K-Fold), Hyperparameter Tuning, Model Evaluation (ROC-AUC, F1, Recall@K), Supervised/ Unsupervised Learning (Random Forest, XGBoost, K-Means, PCA), Multimodal Learning (CLIP, Embeddings, Transformer)

Cloud & Platforms: AWS (EC2, S3, Glue, Athena, RDS, Lambda), Data Lake Architecture, Docker, Git, Jenkins, Streamlit